

SCH 4U FINAL Exam

Part A

MULTIPLE CHOICE

1. For a homogeneous reaction the factor that would NOT affect the reaction rate is:
 - a) surface area of contact
 - b) heterogeneous catalyst
 - c) homogeneous catalyst
 - d) concentrations of reactants
 - e) more than one of the above
2. A temporary structure with a high potential energy that has bonds half formed and half broken is:
 - a) a homogeneous catalyst
 - b) an activated complex
 - c) radical
 - d) an elementary process
 - e) a heterogeneous catalyst
3. The equilibrium constant $K = \frac{[R]^2 [S]}{[P]^3}$ would apply to:
 - a) $2R_{(g)} + S_{(g)} \rightleftharpoons 3P_{(g)}$
 - b) $3P_{(g)} \rightleftharpoons 2R_{(g)} + S_{(g)}$
 - c) $2R_{(s)} + S_{(g)} \rightleftharpoons 3P_{(g)} + P_{(s)}$
 - d) $3P_{(g)} \rightleftharpoons 2R_{(s)} + S_{(g)}$
 - e) more than one of the above
4. Choose the FALSE statement regarding the equilibrium state:
 - a) rate of the reverse reaction = rate of forward reaction
 - b) can occur when ΔG is very large
 - c) all macroscopic properties appear constant
 - d) can only occur in a closed system
 - e) concentrations of all species remain constant
5. The numerical value of the equilibrium constant for a reaction can tell you:
 - a) how long it takes to reach equilibrium
 - b) the extent to which the forward reaction occurs
 - c) whether a reaction is endothermic or exothermic
 - d) whether a temperature increase favors products or reactants
 - e) more than one of the above
6. If K_{sp} for $CuI = 1.27 \times 10^{-12}$. Two solutions are mixed so that the $[Cu^+] = 0.04$ M and the $[I^-] = 0.002$ M in the mixture. The mixture is:
 - a) at equilibrium
 - b) not at equilibrium because the solution is unsaturated
 - c) going to precipitate CuI and therefore will not reach equilibrium
 - d) going to precipitate CuI and then be at equilibrium
 - e) not at equilibrium because the solution is saturated
7. The acid below that would be an Arrhenius Acid but NOT a Bronsted acid is:
 - a) HCl
 - b) HNO_3
 - c) Al^{3+}
 - d) NH_4^+
 - e) none of the above
8. A 0.030 M KOH would have a hydronium ion concentration (in mol/L) of:
 - a) 0
 - b) 0.030
 - c) 3×10^{-9}
 - d) 3×10^{-12}
 - e) 3.33×10^{-13}
9. Solubility of Ag_2CO_3 is 1.3×10^{-4} mol/L. What is the K_{sp} of silver carbonate?
 - a) 8.8×10^{-12}
 - b) 1.7×10^{-8}
 - c) 2.2×10^{-12}
 - d) 3.4×10^{-8}
 - e) 6.8×10^{-8}
10. According to the "Common Ion Effect" compared to the solubility of BaF_2 in pure water, the solubility of BaF_2 will be:
 - a) higher in KNO_3 solution
 - b) higher in HF solution
 - c) higher in $Ba(NO_3)_2$ solution
 - d) lower in NaF solution
 - e) lower in $NaCl$ solution
11. In the reaction: $Cu^{2+} + 4 NH_3 \rightarrow Cu(NH_3)_4^{2+}$ Cu^{2+} acts as a
 - a) Bronsted Base
 - b) Lewis Base
 - c) Bronsted Acid
 - d) Lewis Acid

e) Both a Lewis base and a Bronsted base

12. If a 17.0 mL sample of 0.20 M sulfuric acid (H_2SO_4) is titrated with 0.14 M NaOH, it will require how many mL of titrant to reach neutralization?

- a) 9.7 mL
- b) 48.6 mL
- c) 389 mL
- d) 34.0 mL
- e) 28.6 mL

13. The salt solution below that would be slightly acidic would be:

- a) NaCl
- b) LiCH_3COO
- c) $\text{Cu}(\text{NO}_3)_2$
- d) CaSO_4
- e) LiClO_4

14. Which of the following is a redox reaction:

- a) $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$
- b) $\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2} \text{O}_2$
- c) $\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl}$

d) $2\text{NaI} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{PbI}_2 + 2\text{NaI}$

e) more than one of the above

15. The oxidation number of each boron atom in $\text{B}_4\text{O}_7^{2-}$ is:

- a) 1
- b) 3
- c) 4
- d) -2
- e) 12

16. In the reaction $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

- a) H is the element oxidized
- b) C is the element oxidized
- c) O is the element oxidized
- d) O is increasing its oxidation number
- e) both c and d

17. In the reaction : $\text{Br}_2 + \text{SO}_2 \rightarrow \text{Br}^- + \text{SO}_4^{2-}$
 SO_2 is the

- a) oxidizing agent and gets oxidized
- b) oxidizing agent and gets reduced
- c) reducing agent and gets oxidized
- d) reducing agent and gets reduced

Part B Written Part

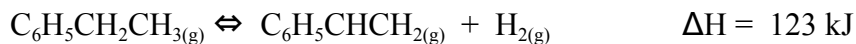
WRITTEN PART: Neatly provide answers to the following questions on the test paper in the space provided. Full marks will only be given for those answers that are organized, detailed and complete.

1. Consider the following reaction:



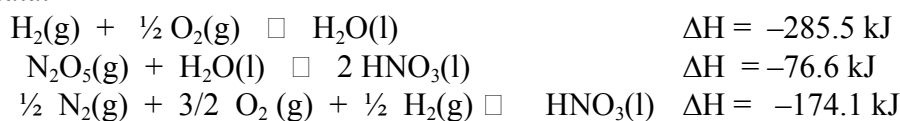
- a. Rewrite the thermochemical equation, including the standard enthalpy of reaction as either a product or a reactant.
- b. Is this exothermic or Endothermic reaction?
- c. Draw an enthalpy diagram for the reaction.
- d. What is the enthalpy change for the formation of 23.6 g of phosphoric acid?

- A sample of ethanol, $\text{C}_2\text{H}_5\text{OH}$, absorbs 23.4 kJ of energy. The temperature of the sample increases from 5.6°C to 19.8°C. What is the mass of the ethanol sample? The specific heat capacity is 2.46 J/g*°C.
- Calculate the specific heat of a metal if a 55.0 g sample of an unknown metal at 99.0 °C causes a 1.7 °C temperature rise when added to 225.0 g of water at 22.0 °C.
- The following equation represents the dissociation of hydrogen iodide gas. At 430° the value of K_c is 0.20.
$$2\text{HI}_{(\text{g})} \rightleftharpoons \text{H}_{2(\text{g})} + \text{I}_{2(\text{g})}$$
Some $\text{HI}_{(\text{g})}$ is placed in a closed container at 430°. Analysis at equilibrium shows that the concentration of $\text{I}_{2(\text{g})}$ is 0.450 mol/L. What are the equilibrium concentrations of $\text{H}_{2(\text{g})}$ and $\text{HI}_{(\text{g})}$?
- Polystyrene resin is manufactured from styrene as follows:



- mixture is heated
- mixture is compressed by changing the volume
- a catalyst is added
- as it forms, the polystyrene resin, $\text{C}_6\text{H}_5\text{CHCH}_2(\text{g})$, further reacts to form the polymer, polystyrene.

6. Given the following data:



calculate the ΔH for the reaction $2 \text{N}_2(\text{g}) + 5 \text{O}_2(\text{g}) \rightarrow 2 \text{N}_2\text{O}_5(\text{g})$ (27.8 kJ)

7. What increase in temperature will result if 212.0g of copper absorbs 4.08kJ of heat energy? Specific Heat Capacity of Copper is $0.385 \text{ J/g}^\circ\text{C}$ (50°C)

8. Write the complete electronic configuration and condensed electronic configuration for the following;

a) U b) Bi

9. Complete the following chart for the following two species.

	FCl_2^+	NO_3^-	PI_5	SO_3
Show your WORK (Need/Have method)				
Final Lewis Diagram (3D)				
Shape				
Bond Angle				
Is this POLAR molecule? Show presence of dipole (if any)				

10. Identify the block, period and group of the following; (a) $3p^5$ and (b) $4d^5$

11. Name the following compound

Draw LINE structure 5-isopropyl-4,6-dimethyloctan-3-ol	Draw CONDENSED structure 1-isopropylbenzene		

12. Draw **LINE structures** of the following

- a) 2-butene b) 2-ethyl-3-methyl-1-butanol c) 3,3-dimethylbutanoic acid
 d) 2,5,5-trimethyl-4-heptone e) 1,8-nonadiyne f) 1,3-diiodobenzene

13. Predict the product, name and draw the product and indicate the type of reaction.

- a. 3-pentyne + (2) water/ H^+ ☐
 b. 2,4-dimethyl-2-hexene + hydrogen bromide ☐
 c. 2-bromo-1-hexanol + benzoic acid ☐

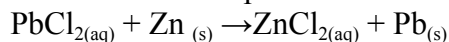
d. 2-methylpent-3-ol + H^+ /heat $\square$

14. The following questions are about this cell: $\text{Al}|\text{Al}^{+3}||\text{Pb}^{+2}|\text{Pb}$
- Identify the anode and the cathode.
 - Write the balanced overall reaction.
 - What is the potential of this cell under standard conditions?

15. Balance this equation: $\text{ClO}_2^- + \text{Br}_2 \square \text{ClO}_4^- + \text{Br}^-$
- In acidic solution
 - In basic solution

16. In an experiment to estimate K_{sp} a piece of Zn metal was left in 50.0 mL of a saturated solution of $PbCl_2$.

The reaction that took place and went to completion was:



The lead that was produced in the displacement reaction was recovered and found to be 0.170 g. Use this information to calculate the K_{sp} for $PbCl_2$. (In your answer, include the solubility equilibrium and the K_{sp} expression for $PbCl_2$.)

17. A student dissolved 5.0 g of vitamin C in 250 mL of water. The molar mass of ascorbic acid (abbreviated HAsc) is 176 g/mol, and its K_a is 8.0×10^{-5} . Calculate the pH of the solution.

18. Consider the titration 20.0 mL of a 0.20 M solution of NaOH with 0.10 M Acetic ACID (CH_3CH_2COOH). Calculate:

- The pH before any titrant has been added.
- The pH after 10.0 mL of titrant (base) has been added.
- The pH at the equivalence point
- What volume corresponds to a position half-way to the equivalence point? What is the significance of this point? ($pH = pK_a$)